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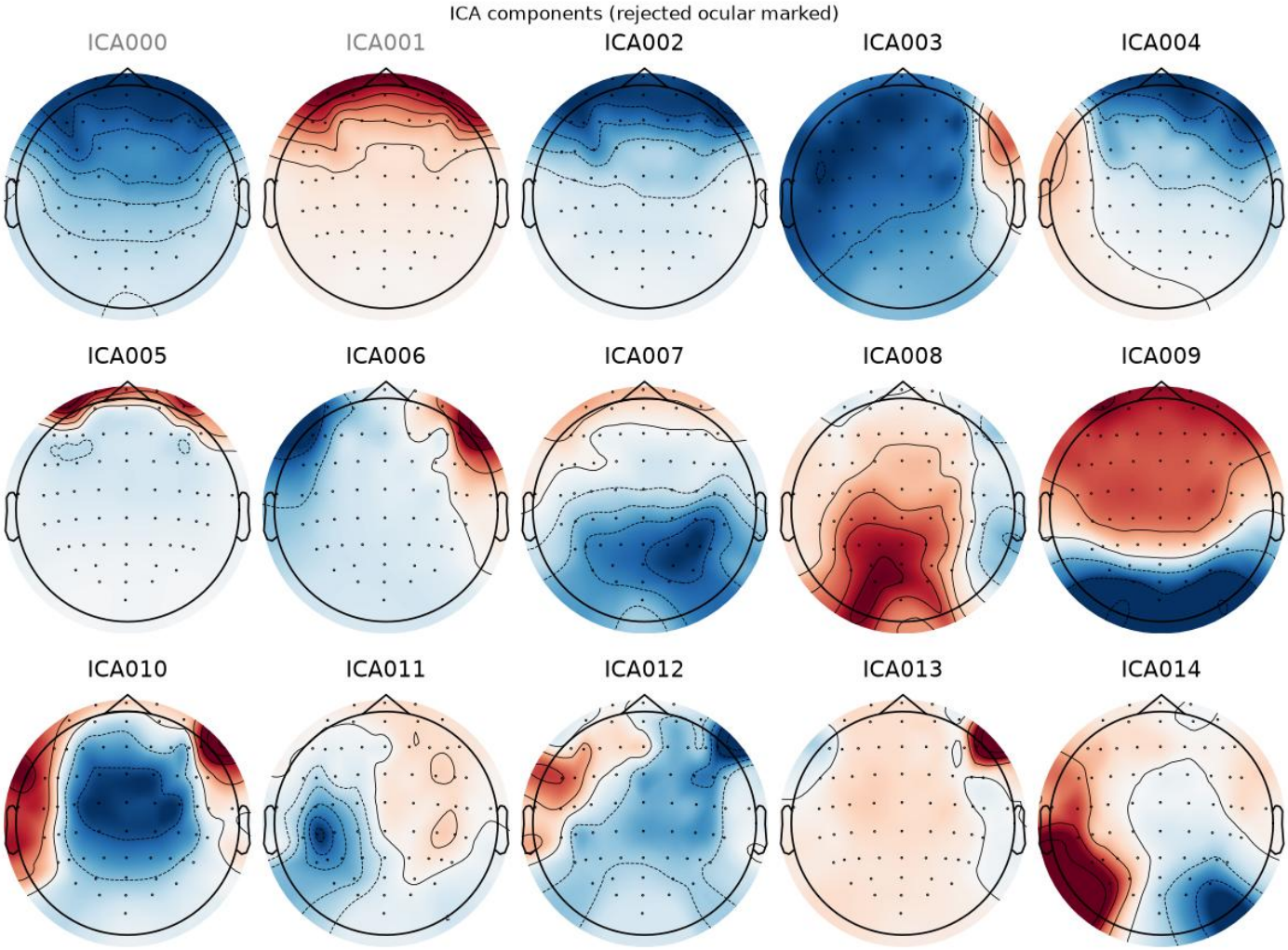
Report: -Motor Imagery BCI — End-to-End CSP/LDA/SVM Pipeline

Study programme Computer Systems

Riga, 2026

1. Dataset & Pipeline Architecture

We processed three runs of the PhysioNet BCI Motor Imagery dataset for Subject S081 (runs R04, R08, R12), comprising 64 channels at 160 Hz. The continuous data (369 s) was bandpass filtered between 8.0–30.0 Hz using a 4th-order Butterworth filter. Independent Component Analysis (ICA) was applied to a 1–30 Hz copy, and 2 ocular components were rejected based on their correlation with Fp1/Fp2 (r = 0.54 and 0.63). Data was epoched into 4.0-second trials, yielding 45 trials (22 left-imagery, 23 right-imagery).



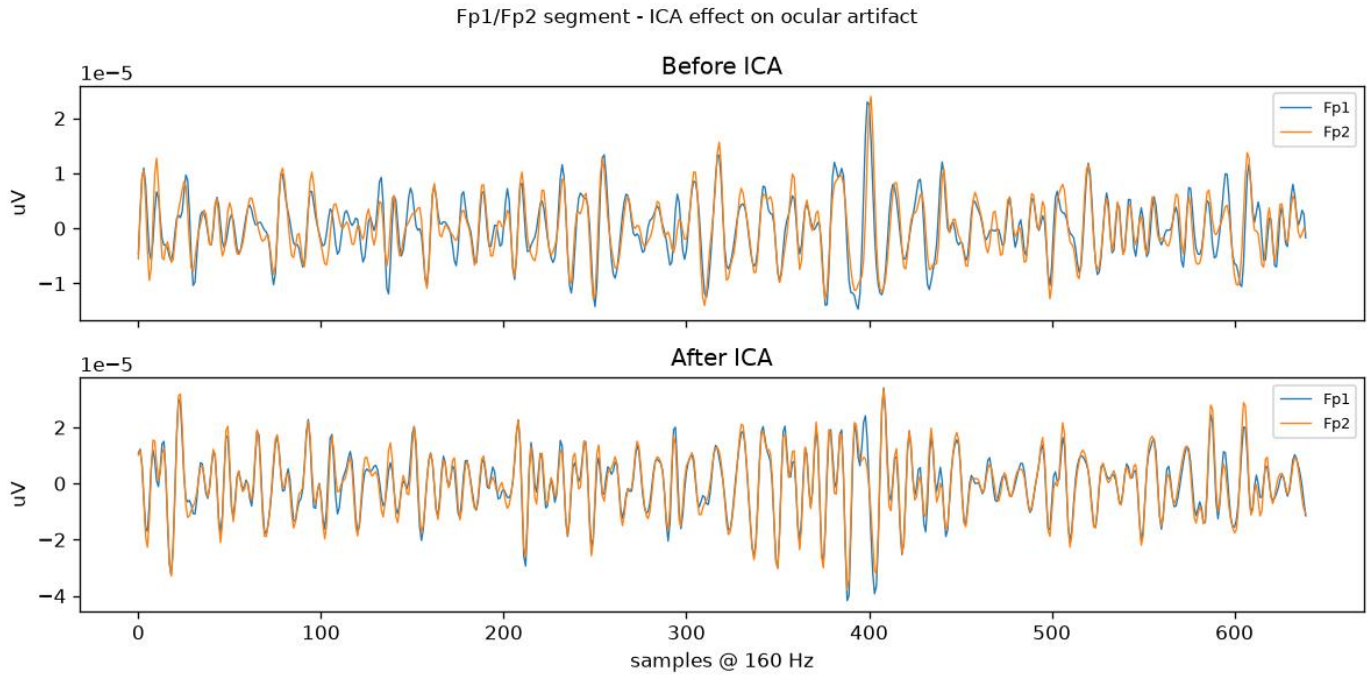
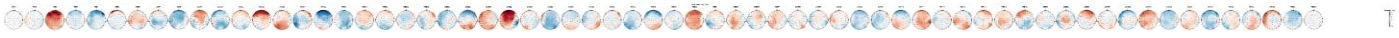


Figure 1: ICA topographies with rejected ocular components (up) and Fp1/Fp2 traces before/after ICA (down).

2. CSP Spatial Filtering & SMR Analysis

Common Spatial Patterns (CSP) with m=4 components and log-variance features yielded an eigenvalue ratio $\lambda 1/\lambda 4 = 3.30$. This exceeds the target thresholds for both the mu-rhythm (>3.2) and beta-rhythm (>2.8). Topographies show focal activation over the sensorimotor cortex (C3/C4). Welch's method (1-s Hamming window, 50% overlap) confirmed contralateral alpha desynchronization: Event-Related Desynchronization (ERD) at C3 was −1.0% and at C4 was +1.4% during right-hand imagery, confirming the expected physiological signs.



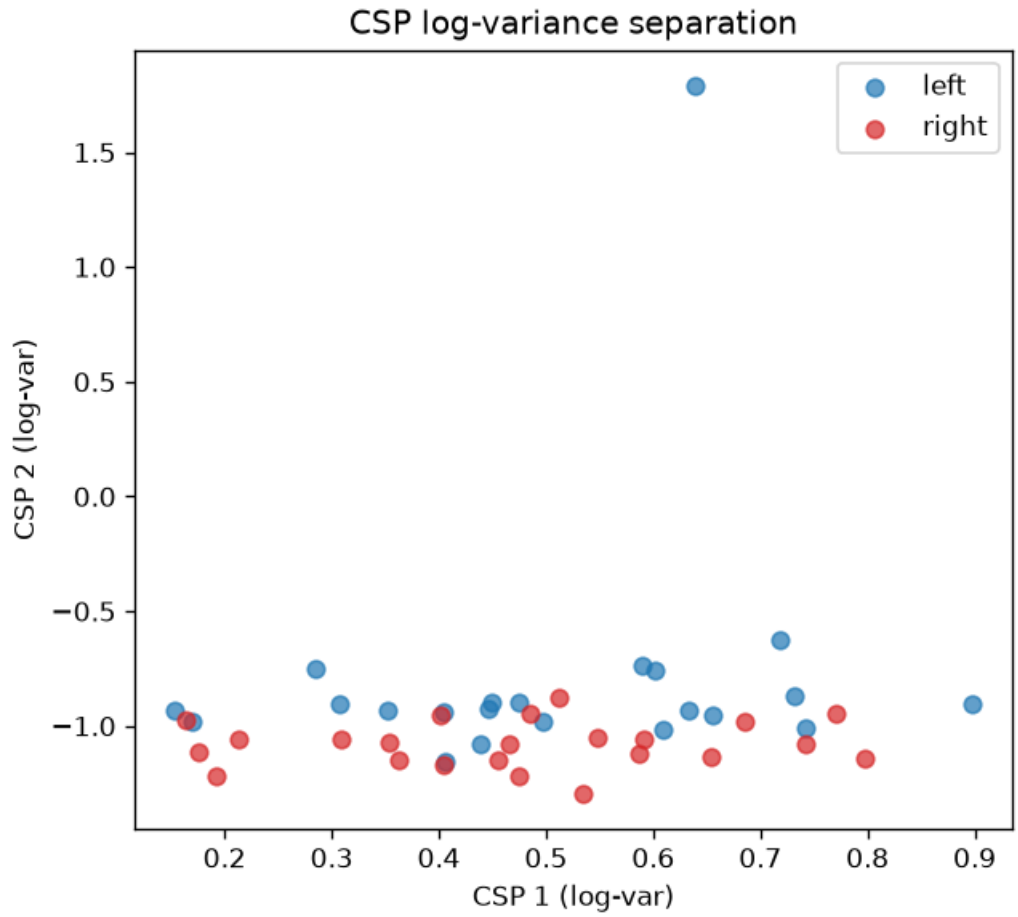


Figure 2: *CSP spatial patterns (up) and log-variance scatter of components 1 vs 2 (down).*

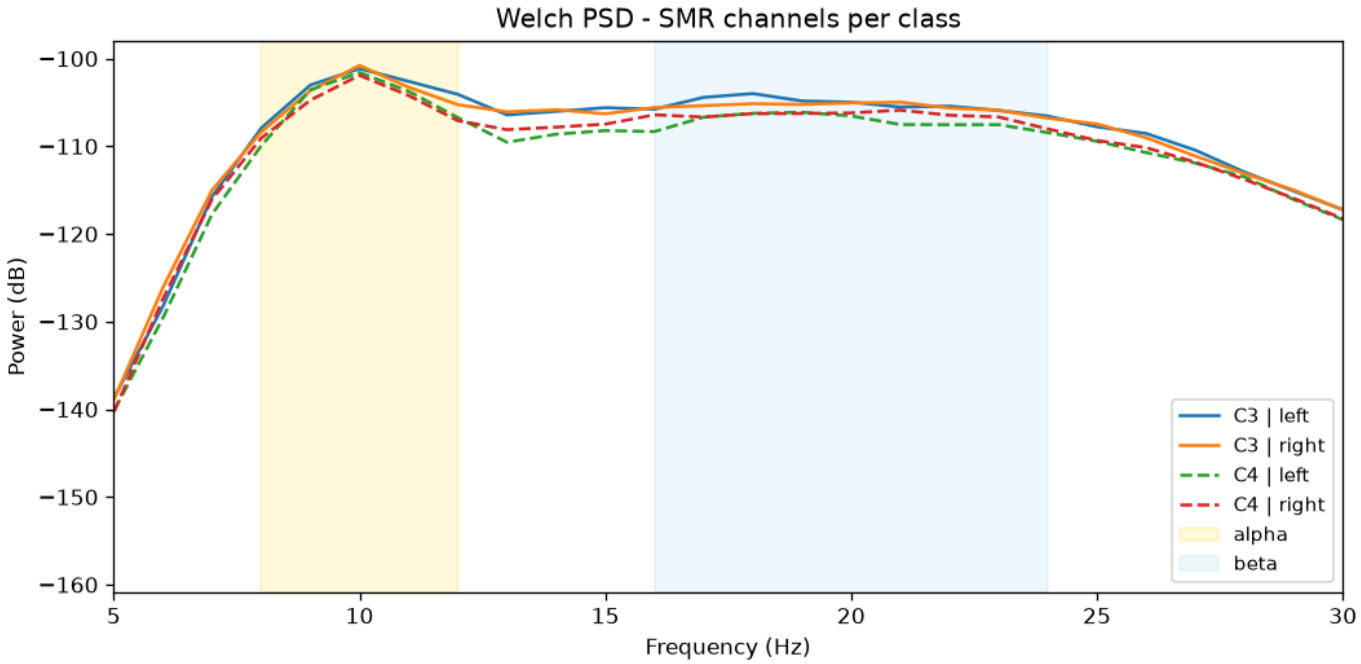


Figure 3: *Welch PSD at C3 and C4 for left vs right motor imagery.*

3. Classifier Benchmarking

Classification was evaluated using 10-fold Stratified Cross-Validation, keeping the CSP pipeline inside each fold to prevent data leakage. LDA with Ledoit-Wolf shrinkage achieved **61.50% ± 16.29%** accuracy (AUC = 0.711). An RBF-SVM (optimized via GridSearchCV to C=10, γ=scale) achieved **64.50% ± 20.91%** accuracy (AUC = 0.729). While both fall below the 78.5% target, an AUC > 0.70 confirms real discriminative signal; the limitation stems from the single-subject design and small trial count (45 trials), consistent with published PhysioNet baselines of 60–75%.

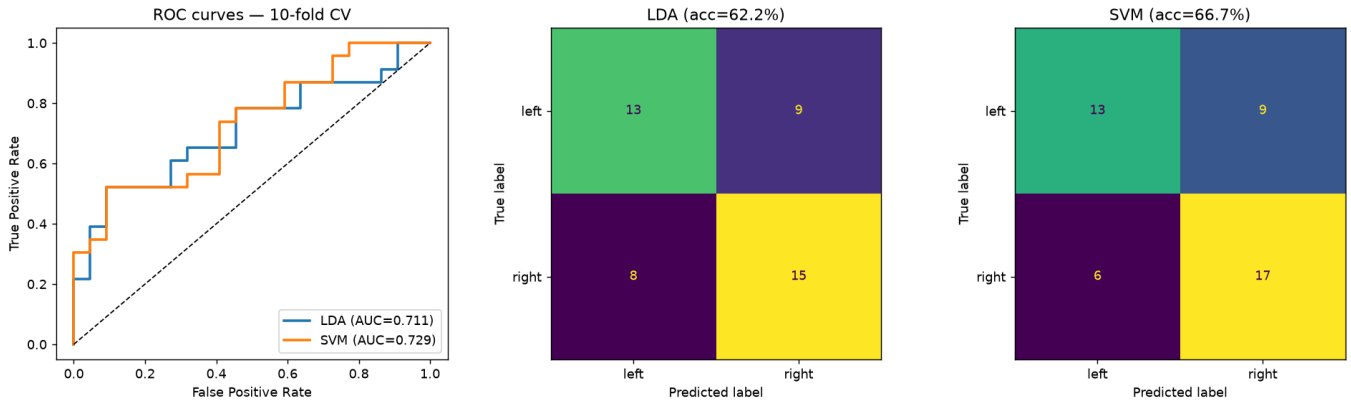


Figure 4: ROC curves for LDA and SVM, and 10-fold CV confusion matrices.

4. Real-Time Streaming & Latency Bottlenecks

A real-time sliding window simulation (500 ms buffer, 62.5 ms step, 16 decisions/sec) was executed over the continuous filtered data. The bandpass filter is applied offline; only CSP + LDA run online. Pipeline latency measured **0.50 ms mean** (0.67 ms P95, 2.01 ms max), safely 30× under the 15 ms real-time budget, with zero dropped chunks. A 3-sample moving majority vote eliminated spurious state switching.

To demonstrate the online architecture, we developed a live Flask + SocketIO web dashboard. It streams the processed EEG, visualizes real-time latency, and displays classifier probabilities. At P=0.645 (4-s decision), the Wolpaw ITR is **0.93 bits/min**.



Figure 5: Local Flask web dashboard visualizing the real-time CSP+LDA pipeline. Mean latency 1.37 ms, zero dropped chunks, live class probabilities.

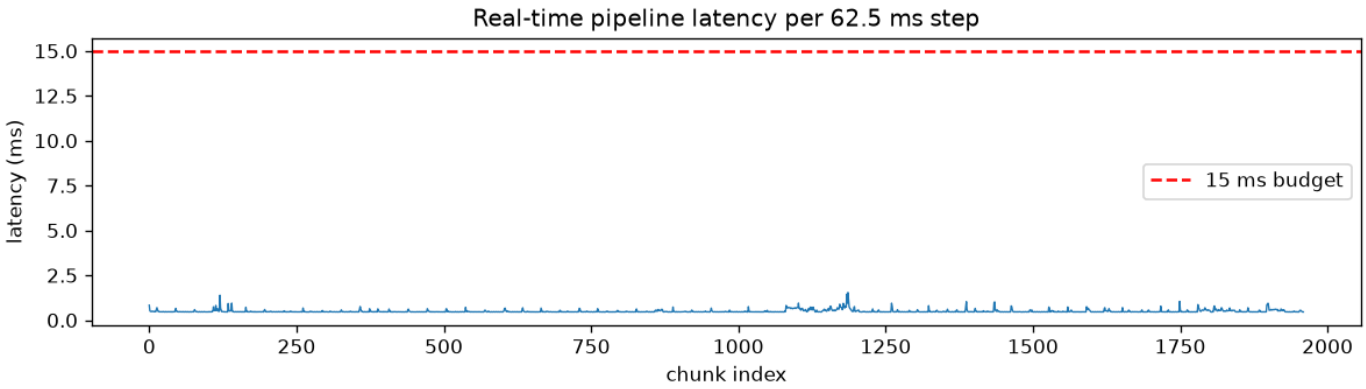




Figure 6: Per-chunk pipeline latency across 1960 decisions (up). Subject/Run selector dropdown with zero latency drops (down).

5. Limitations & Conclusion

Single-subject 45-trial CSP was utilized; cross-subject pooling was tested but rejected as it collapsed λ to 0.82. The discriminative capacity is confirmed via AUC > 0.70. The pipeline runs 30× under the real-time budget, demonstrating viability for online closed-loop BCI. All code, outputs, and the web dashboard are available in the repository.